



Title

kdrobust — Kernel Density Estimation with Robust Bias-Corrected Confidence Intervals and Inference Procedures.

Syntax

```
kdrobust varname [if] [in] [, eval(gridvar) neval(#) h(#) b(#) rho(#)
kernel(kernelfn) bwselect(bwmethod) bwcheck(#) imsegrid(#) level(#)
separator(#) genvars plot graph_options(gphopts) ]
```

Description

kdrobust implements kernel density point estimators with robust bias-corrected confidence intervals and inference procedures developed in [Calonico, Cattaneo and Farrell \(2018\)](#). See also [Calonico, Cattaneo and Farrell \(2022\)](#) for related optimality results. It also implements other estimation and inference procedures available in the literature. See Wand and Jones (1995) and Fan and Gijbels (1996) for background references.

A detailed introduction to this command is given in [Calonico, Cattaneo and Farrell \(2019\)](#).

Companion command is: [kdbwselect](#) for data-driven bandwidth selection.

Related Stata and R packages useful for empirical analysis are described in the following website:

<https://nppackages.github.io/>

Options

eval(gridvar) specifies the grid of evaluation points for *xvar*. By default it uses 30 equally spaced points over to support of *xvar*.

neval(#) specifies the number of evaluation points to estimate the regression functions. Default is 30 evaluation points.

h(hvar) specifies the main bandwidth (*h*) used to construct the point estimator for each evaluation point. If not specified, it is computed by the companion command [kdbwselect](#).

b(bvar) specifies the bias bandwidth (*b*) used to construct the bias-correction estimator for each evaluation point. If not specified, it is computed by the companion command [kdbwselect](#).

rho(#) specifies the value of *rho*, so that the bias bandwidth *b* equals $b=h/\rho$. Default is **rho(1)** if *h* is specified but *b* is not.

kernel(kernelfn) specifies the kernel function used to construct the kernel density estimator(s). Options are: **epanechnikov**, and **uniform**. Default is **kernel(epanechnikov)**.

bwselect(bwmethod) bandwidth selection procedure to be used. By default it computes both *h* and *b*, unless *rho* is specified, in which case it only computes *h* and sets $b=h/\rho$. Options are:

mse-dpi second-generation DPI implementation of MSE-optimal bandwidth. Default choice.

mse-rot ROT implementation of MSE-optimal bandwidth.

imse-dpi second-generation DPI implementation of IMSE-optimal bandwidth.

imse-rot ROT implementation of IMSE-optimal bandwidth.

ce-dpi second generation DPI implementation of CE-optimal bandwidth.

ce-rot ROT implementation of CE-optimal bandwidth.

Note: MSE = Mean Square Error; IMSE = Integrated Mean Squared Error; CE = Coverage Error; DPI = Direct Plug-in; ROT = Rule-of-Thumb.

Default is **bwselect(mse-dpi)**. For details on implementation see [Calonico, Cattaneo and Farrell \(2019\)](#).

bwcheck(#) specifies an optional positive integer so that the selected bandwidth is enlarged to have at least # effective observations available for each evaluation point.

imsegrid(#) number of evaluations points used to compute the IMSE bandwidth selector. Default is 30 points.

level(#) specifies confidence level for confidence intervals. Default is **level(95)**.

separator(#) draws separator line after every # variables; default is **separator(5)**.

plot generates the local polynomial regression plot.

genvars generates new variables storing the following results.

- kdrobust_eval** evaluation points.
- kdrobust_h** bandwidth h.
- kdrobust_b** bandwidth b.
- kdrobust_nh** effective sample size.
- kdrobust_gx_us** conventional local polynomial estimate.
- kdrobust_se_us** conventional standard error for the local polynomial estimator.
- kdrobust_gx_bc** bias-corrected local polynomial regression estimate.
- kdrobust_se_rb** robust standard error for the local polynomial estimator.
- kdrobust_ci_l_rb** lower end value of the robust confidence interval.
- kdrobust_ci_r_rb** upper end value of the robust confidence interval.

graph_options(*gphopts*) specifies graphical options to be passed on to the underlying graph command.

Setup

. **sysuse auto**

Kernel density estimates for length

. **kdrobust length**

Kernel density estimates for length

. **kdrobust length, plot genvars**

Saved results

kdrobust saves the following in **e()**:

Scalars

e(N) original number of observations

Macros

e(varname) name of variable
e(bwselect) bandwidth selection choice
e(kernel) kernel choice

Matrices

e(Result) estimation result

References

- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2018. On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference. *Journal of the American Statistical Association*, 113(522): 767-779.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2019. nprobust: Nonparametric Kernel-Based Estimation and Robust Bias-Corrected Inference. *Journal of Statistical Software*, 91(8): 1-33. doi: 10.18637/jss.v091.i08.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2022. Coverage Error Optimal Confidence Intervals for Local Polynomial Regression, *Bernoulli*, forthcoming.

Fan, J., and Gijbels, I. 1996. Local Polynomial Modelling and Its Applications, London: Chapman and Hall.

Wand, M., and Jones, M. 1995. Kernel Smoothing, Florida: Chapman & Hall/CRC.

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